Telecommunications Policy 49 (2025) 102976



Contents lists available at ScienceDirect

Telecommunications Policy

journal homepage: www.elsevier.com/locate/telpol





# Transforming cybersecurity with agentic AI to combat emerging cyber threats

Nir Kshetri

*Bryan School of Business and Economics, The University of North Carolina at Greensboro, Bryan Building, Room: 368, P. O. Box 26165, Greensboro, NC, 27402-6165, USA*

A B S T R A C T

This paper investigates the transformative potential of agentic AI in cybersecurity, specifically addressing how it can enhance practices in response to emerging threats. It aims to explore how agentic AI can transform cybersecurity practices, particularly in addressing new and evolving threats, while also examining the cybersecurity risks associated with its integration. The research explores the possibilities for agentic AI to automate critical tasks within Security Operations Centers (SOCs), such as decision-making, incident response, and threat detection. It also emphasizes the risks associated with AI integration, including the introduction of new vulnerabilities and challenges in managing automated systems, which call for a reassessment of existing cybersecurity frameworks to effectively address these risks.

## 1. Introduction

In 2024, the global artificial intelligence (AI) in cybersecurity market was valued at $24.8 billion. It is expected to grow to $29.6 billion in 2025 and exceed $146.5 billion by 2034 (Precedence Research, 2024). Agentic AI, representing the third wave of AI, is expected to account for a growing share of this market. Agentic AI strengthens Security Operations Centers (SOCs) by analyzing large datasets of threat intelligence and improving the detection of real threats versus false positives. Large language models (LLMs) trained for this purpose offer real-time, contextual insights, saving analysts time (Columbus, 2025). Agentic AI systems can autonomously detect, respond to, and mitigate security and fraud threats in near real-time (CIO, 2024). AI agents enable personalized protocols that adapt to specific threats and vulnerabilities (CIO, 2024). A network of specialized models, each tasked with identifying different vulnerabilities, share insights, and collectively address threats (Institute for Experiential AI., 2025). For example, cybersecurity company Darktrace's agentic AI, Enterprise Immune System, detects and responds to threats by learning normal network behavior. Rather than relying on security teams, it autonomously mitigates risks by quarantining suspicious devices or blocking unusual data transfers in real time (Hoque, 2025). This capability reduces response times to potential attacks as well as costs, strengthening overall security by providing faster, more efficient threat management (Castellanos, 2021) and enhancing system resilience (Weigand, 2025). This streamlines workflows, minimizes manual intervention (CIO, 2024), enhances SOC efficiency, and enables faster decision-making (Columbus, 2025). Research and consulting firm Gartner's report, *Predict 2025: There Will Never Be an Autonomous SOC*, forecasts that AI will enhance SOC efficiency by 40 % by 2026, compared to 2024. This increase in efficiency will shift SOC roles toward AI development, maintenance, and security, marking a significant change in how cybersecurity teams operate (Columbus, 2025).

AI is this soon expected to shift from chatbots to agent-driven cybersecurity, enhancing threat detection, autonomous responses, scalability, and cyber hygiene (Weigand, 2025). As AI systems security teams may delegate tasks like tracking supply-chain weaknesses, maintaining system inventories, and handling alerts, improving overall defense capabilities (Klappholz, 2025). Given the global shortage of nearly 4 million cybersecurity professionals (World Economic Forum, 2024), agentic AI has the potential to fill this

*E-mail address:* nbkshetr@uncg.edu.

https://doi.org/10.1016/j.telpol.2025.102976
Received 26 February 2025; Received in revised form 18 April 2025; Accepted 18 April 2025
Available online 22 April 2025
0308-5961/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

gap by automating tasks, improving efficiency, and enhancing security operations. Analysts also predict the growing prevalence of virtual CISO (vCISO) roles as organizations face challenges in finding and retaining top cybersecurity talent. Many businesses may turn to vCISO services to access high-level expertise on a flexible basis, which could democratize access to experienced cybersecurity professionals. This trend enables smaller organizations to benefit from expert guidance without the cost of hiring a full-time CISO (Right-Hand AI., 2025a,b).

A KPMG survey of UK tech leaders found that half of the boards highlighted cybersecurity issues as a significant worry when adopting generative AI tools (Woollacott, 2024). Adversaries are achieving breakout times of just 2 min and 7 s, highlighting the need for faster detection and response in cybersecurity (Columbus, 2025). The various benefits of agentic AI mentioned earlier are expected to help mitigate cybersecurity concerns.

On the other side of the coin, with any new technology, existing threats persist, new threats emerge, and traditional vulnerabilities become more consequential (Baker, 2025). Like other emerging technologies (e.g., Kshetri, 2021), agentic AI can also bring increased security risks. Unlike traditional AI systems that function within controlled environments, AI agents interact with various systems and external data sources, expanding the potential attack surface. This can lead to unauthorized access, data leakage, and other vulnerabilities. Weak integration or system flaws have previously allowed sensitive data to be exposed, while malicious actors or coding errors can manipulate AI agents, causing unintended disruptions or financial losses (Ramesh, 2025). If an agentic AI system is hacked, the consequences can be severe. First, detecting and confirming the breach may take time, and even a minor change can lead to significant effects. Second, the system's autonomy introduces new security vulnerabilities that must be addressed. Finally, the greatest risk lies in deploying a system without proper monitoring, logging, and controls—these cannot be added as afterthoughts (Kaufman, 2024).

This paper analyzes how agentic AI can transform cybersecurity practices while also introducing security risks, addressing two main research questions.

**RQ1**. How can agentic AI transform cybersecurity practices in addressing emerging threats?;

**RQ2**. What are the cybersecurity risks associated with agentic AI?

The paper is structured as follows. First, we introduce the concept of AI agents and agentic AI, discussing their functionality, evolution, and key differences between traditional AI and agentic AI. Next, we explore how agentic AI enhances cybersecurity, highlighting its applications and benefits in securing digital systems. Following this, we examine the security and privacy risks associated with AI agents, proposing a Risk Management Framework to address these challenges. We then analyze the malicious potential of agentic AI in cybercrime, exploring how it could be exploited for harmful purposes. Afterward, we discuss the broader implications of agentic AI on the cybersecurity landscape and the necessary steps for regulation. Finally, we offer concluding remarks.

## 2. Understanding AI agents and agentic AI: functionality, evolution, and key differences

In this section, we introduce AI agents and agentic AI, highlighting their ability to autonomously perceive, adapt, and act, which plays a crucial role in shaping both cybercrime tactics and cybersecurity defenses. An AI agent is an intelligent system (Boston Consulting Group, 2025) or a smart digital assistant that uses artificial intelligence techniques—like machine learning, natural language processing, or reinforcement learning—to learn from data, make decisions, and perform tasks on its own without constant human input (Carter, 2025), in order to achieve specific goals (Boston Consulting Group, 2025). AI agents can retain memory across tasks, integrate multiple AI models to execute complex actions, and determine when to access internal or external systems autonomously. This allows them to make decisions and perform actions with minimal human intervention, streamlining processes and improving efficiency (Boston Consulting Group, 2025).

A consumer goods company used an AI agent to optimize global marketing campaigns, reducing the workload from six analysts weekly to one employee with results in under an hour. The AI agent autonomously gathers marketing data, analyzes performance metrics, and drafts optimization recommendations. After refinement and approval by an operator, it updates media buying platforms with the proposed changes (Boston Consulting Group, 2025).

AI agents operate through several interconnected modules to ensure effective functionality. The Interfaces enable communication with users, systems, and databases. The Memory Module stores both short-term context and long-term knowledge from past interactions. The Profile Module outlines the agent's goals, roles, and behaviors. The Planning Module uses AI models to develop actionable plans. Finally, the Action Module integrates APIs to carry out these plans. Together, these components allow AI agents to interact, learn, and act within their environments (Boston Consulting Group, 2025).

AI agents originated in the 1990s as autonomous software designed to navigate simple internet environments on behalf of users. With limited search engines and basic web content, AI agents were seen as useful for tasks like browsing, as the technology was not yet capable of handling the complexities of the real world. The rise of LLMs has spurred renewed interest in AI agents. Companies like Salesforce and ServiceNow are using specialized agents to automate tasks, analyze data, and personalize interactions. These agents leverage LLMs to understand natural language and determine when to call tools like web searches, code execution, and data retrieval commands, enhancing operational efficiency (Hodjat, 2025).

While both AI agents and agentic AI function autonomously, their capabilities differ significantly. AI agents are task-specific, operating within predefined rules to handle routine activities such as scheduling meetings or responding to customer queries. In contrast, agentic AI demonstrates a higher degree of autonomy, capable of navigating complex environments and pursuing long-term goals. It employs adaptive reasoning and iterative planning, enabling it to learn from experience and adjust its strategies dynamically

(CyberArk, n.d.). Agentic AI continuously learns from experience and data, whereas AI agents often require reprogramming. Additionally, agentic AI integrates with various technologies, actively perceives its environment, and proactively responds to changes, while AI agents are more limited and reactive in scope. Overall, agentic AI is designed for dynamic, autonomous tasks, while AI agents excel in narrower, defined functions (Nawaz, 2024).

Agentic AI systems are also described as advanced AI agents or AI super-agents, which can autonomously perform complex tasks and make decisions (Dorsey & Whitney, 2025). An agentic framework is a collection of capabilities, such as LLMs and memory, combined to create an AI agent or solution.

Agentic AI can be a physical robot, software, or a combination of both, controlling and interacting with other software tools. In robotic systems, it uses sensors and cameras to gather data and makes decisions based on this information through sensor fusion. In software, agentic AI collects data from various sources like APIs, databases, and online searches to form context and perception for decision-making (Red Hat. 2024).

## 3. Enhancing cybersecurity with agentic AI and AI agents

Agentic AI is revolutionizing SOCs by automating decision-making and adapting to evolving threats (Columbus, 2025). Agentic AI enhances workflow efficiency by automating tasks like alert triage and incident response, with applications in cybersecurity for autonomous threat detection and response (Lisowski, 2024). Organizations adopting agentic AI for security and governance are reporting significant benefits, including faster decision-making, and improved risk management (Chiodi, 2025).

AI systems enable dynamic access management by adjusting user access based on behavior and risk, aligning with zero trust principles. Additionally, agentic AI continuously monitors activities, identifying anomalous patterns to detect and mitigate insider threats before they escalate. Furthermore, it helps reduce the costs associated with cybersecurity incidents caused by disconnected applications, a major source of breaches. By integrating these applications into existing identity and governance platforms, agentic AI enhances overall security and minimizes financial losses from such incidents. (Chiodi, 2025).

In addition to advancements in cybersecurity, agentic AI can also enhance software development by streamlining processes throughout the software development lifecycle (SDLC). Integrating intelligent agents into the SDLC enables organizations to shift from reactive to proactive AppSec practices. These AI-powered agents continuously analyze code repositories for vulnerabilities using techniques like static code analysis and machine learning. What sets agentic AI apart is its ability to adapt to each app's unique structure, building a comprehensive code property graph (CPG) to identify security flaws based on their impact, rather than relying on generic severity ratings (Guerrero, 2025). A CPG is a flexible, language-neutral representation of program code that supports scalable, incremental, and distributed analysis (ShiftLeftSecurity, 2025). The most intriguing use of agentic AI in AppSec is automated vulnerability fixing. Traditionally, human programmers manually identify and fix vulnerabilities—a process that is time-consuming and error-prone. With agentic AI and Code Property Graph analysis, AI agents can detect vulnerabilities and generate context-aware, non-breaking fixes that address security flaws without introducing new bugs (Guerrero, 2025). Agentic AI systems are less susceptible to "hallucinations" or errors commonly seen in generative AI, as they are trained on specific tasks, leading to improved accuracy and reliability (Chiodi, 2025).

### *3.1. Agentic AI solutions revolutionizing cybersecurity providers*

Agentic AI is emerging as a game-changer in cybersecurity, with predictions pointing to its dominance by 2026. Microsoft CEO Satya Nadella envisions these AI systems managing multiple SaaS applications, orchestrating sophisticated security measures. By running continuous simulations and adapting defenses in real time, agentic AI will establish a dynamic, interconnected security ecosystem. Tapping into global intelligence sources like Interpol and dark web insights, it will proactively counter cyber threats and promote international collaboration through initiatives such as Cyber Eagle (Tortora Brayda, 2025). Leading cybersecurity providers have started offering agentic AI solutions to enhance efficiency and strengthen security (Columbus, 2025).

Table 1 provides an overview of various AI agents and platforms utilized by cybersecurity companies, highlighting their performance in enhancing security detection, automating workflows, and reducing response times.

**Table 1**
Some AI agents and their cybersecurity performance.

| Company | AI agent/platform | Cybersecurity performance |
|---|---|---|
| CrowdStrike | Charlotte AI Detection Triage | Delivers over 98 % accuracy in security detection triage, automating over 40 h of manual work weekly to improve SOC operations and speed up threat responses. |
| ReliaQuest | GreyMatter platform | Processes security alerts 20 times faster than traditional methods.<br>Automates 98 % of security alerts<br>Reduces threat containment time to under 5 min. |
| Twine | "Digital employee" Alex | Identifies vulnerabilities and proactively prevents unauthorized access, easing the workload for IT and cybersecurity team. |
| Darktrace | Darktrace/ENDPOINT | Reduces triage time by 90 % (Darktrace, 2024a, 2024b). |
| Microsoft | Security Copilot featuring 11 task-specific AI agents | 30 % reduction in mean time to respond (MTTR) to security incidents.<br>Early-career professionals: 26 % faster and 35 % more accurate in threat handling.<br>Experienced professionals: 22 % faster and 7 % more accurate (Barker, 2025). |

### 3.2. ReliaQuest's GreyMatter platform

Cybersecurity technology company ReliaQuest claimed its autonomous AI security agent, launched in September 2024, processes security alerts 20 times faster than traditional methods while improving threat detection accuracy by 30 %. This advancement could significantly enhance cybersecurity response times and efficiency (Klappholz, 2025). The autonomous AI agent for security operations automates 98 % of security alerts and reduces threat containment time to under 5 min. The system uses multiple agents and models to execute various tasks simultaneously (Williams, 2025). ReliaQuest claimed its AI agent, trained on over a decade of incident response data, autonomously handles tier 1 and tier 2 security tasks by analyzing alerts and executing necessary actions, using real-time, customer-specific data in a secure, private environment to eliminate AI hallucinations (Williams, 2025). The system was reported to offer full transparency for security teams to monitor decision-making and prioritizes data privacy by operating in segmented, context-specific environments, ensuring protection from public models (Williams, 2025). As of early 2025, it was deployed across 1000+ enterprises (Williams, 2025).

### 3.3. CrowdStrike Falcon Agent

Designed to integrate various advanced endpoint protection features, the CrowdStrike Falcon Agent—integral to the CrowdStrike Falcon platform—operates with a streamlined, lightweight architecture under 20 MB (Cosive, 2025). The platform, driven by AI and the CrowdStrike Security Cloud, monitors attack indicators, threat intelligence, and telemetry data to deliver accurate threat identification, automated protection and remediation, advanced threat hunting, and vulnerability prioritization (CrowdStrike, 2025a,b). CrowdStrike's Charlotte AI Detection Triage, launched in February 2025, triages security detections with over 98 % accuracy, eliminating more than 40 h of manual work per week, and enhancing SOC operations and response times (CrowdStrike, 2025a,b).

Among the vendors driving the development of agentic AI in cybersecurity, CrowdStrike stands out as a leader in terms of both adoption and innovation. Its agent, Charlotte AI, is uniquely trained on proprietary data from its Falcon Complete platform, allowing for precise threat detection based on real-world adversary tactics (CrowdStrike, 2025a,b). In partnership with Nvidia, CrowdStrike has also integrated LLama Nemotron AI models and NIM microservices into Charlotte AI. This enhances the Falcon platform's automated threat detection while reducing GPU demands. The collaboration addresses a key industry pain point: overloaded security teams. By embedding advanced AI into security operations, the companies have enabled faster threat response, minimized false positives, and improved analyst productivity. These developments exemplify how agentic AI is not only transforming cybersecurity workflows but also becoming a strategic differentiator for vendors. As High (2025) observes, the future of cybersecurity hinges on AI systems capable of advanced reasoning and real-time collaboration with human analysts.

### 3.4. Twine's "digital employee" Alex

Twine, a Tel Aviv, Israel-based cybersecurity startup, aims to address the talent shortage by developing AI agents like Alex, its first "digital employee," specializing in identity and access management (IAM). In this role, Alex performs the task of identifying vulnerabilities and taking proactive steps to block unauthorized access, reducing the burden on IT and cybersecurity teams. (Riley, 2024). Deployed as a SaaS platform, Alex connects to customer systems, creates task plans, seeks approval, and executes them with full visibility. In November 2024, Twine $12 million in seed funding. Twine is the first company to pioneer cybersecurity digital employees, building on earlier models like Amelia by IPsoft (Press, 2024).

### 3.5. Darktrace/ENDPOINT

Using Agentic AI, Darktrace autonomously monitors, detects, and mitigates potential cyber risks as they occur, ensuring continuous learning and protection (Lisowski, 2024). Darktrace/ENDPOINT uses lightweight agents to analyze datapoints from every network packet and connection to uncover unusual activity in real-time, including remote devices or if users are off-VPN. The self-learning AI is deployed locally, trained exclusively on companies' data without requiring a cloud connection, providing tailored security outcomes while maintaining privacy. Darktrace/ENDPOINT detects threats by learning normal network behavior, identifying both known and novel threats without relying on external data. It continuously analyzes endpoint connections, complementing existing endpoint detection and response (EDR) solutions to uncover internal and external risks, including zero-days, supply chain attacks, and insider threats (Darktrace, 2024a, 2024b).

### 3.6. Microsoft

In March 2025, Microsoft announced 11 Security Copilot agents powered by AI to accelerate threat response—six created by Microsoft and five by partners, with one specifically integrated into Microsoft Purview for Data Security Investigations (DSI). Microsoft's tools include phishing and alert triage (Defender, Purview), identity policy optimization (Entra), vulnerability prioritization (Intune), and threat intelligence curation. Partners like OneTrust, Aviatrix, BlueVoyant, Tanium, and Fletch contributed agents for breach reporting, network diagnostics, SOC control assessments, alert prioritization, and threat forecasting. The Purview DSI agent tackles data exposure risks (Barker, 2025).

## 4. Security and privacy risks of AI agents: Risk Management Framework

As agentic AI adoption grows, with bots autonomously acting on users' behalf, traditional methods of identifying malicious automation will become outdated. Security systems must transition from detecting automation to predicting behavior and intent, requiring a different risk management approach and introducing new challenges in API security and bot management (Weigand, 2025).

The Risk Management Framework (RMF) emphasizes the idea that cybersecurity is about effectively managing risks rather than completely eliminating threats. Organizations using RMF assess risks by considering threats, vulnerabilities, and the potential impact on their operations. Based on this assessment, they implement appropriate security measures to mitigate risks, ensuring they maintain a secure environment while complying with industry standards and regulations (NIST, 2024). It is argued that there is no "cyber-alchemy formula" to accurately determine the best cyber tools and optimal level of cybersecurity spending. Practitioners have advocated for a process-based model, where the formula is based on a risk-based strategy, defined as "threat plus vulnerability plus consequences"" (Finch, 2014) (Table 2).

### *4.1. Threat*

A cyberthreat refers to any potential risk or danger that could compromise the security or integrity of an organization's digital systems, data, or network infrastructure (Kshetri, 2021), especially when agentic AI is involved. The many features and capabilities of agentic AI bring potential threats, as they can expose sensitive systems, data, and networks to cyber-attacks, increasing the risk of harm or disruption. Agentic LLMs pose significant data security risks due to their deep access to sensitive information and system functionalities, making them prime targets for cyberattacks (SC World, 2025).

The potential threats to agentic AI can be analyzed by examining its attack surface. An attack surface consists of all potential points where unauthorized users can attempt to access or extract data from a system, including exposed and vulnerable software, network, and hardware points (The Hacker News., 2025). Agentic AI increases the attack surface for cybercriminals and other offenders. Unlike standard AI models, where threats are limited to inputs, processing, outputs, and software vulnerabilities, AI agents expand the attack surface to include the entire chain of interactions they initiate—many of which remain invisible to human or system operators (Vella, 2024). For instance, companies may face challenges with "shadow AI," where employees deploy agentic AI tools, such as public AI models for data analysis or AI-powered coding assistants like GitHub Copilot. without proper authorization and sanction. While these tools can increase productivity, they can also create significant security and privacy risks if not properly managed (Weigand, 2025). For instance, an employee may unknowingly input sensitive company data into a public AI tool, leading to potential data breaches. Similarly, when using AI coding assistants without proper oversight, employees might inadvertently introduce vulnerabilities into the organization's software, exposing the system to potential cyberattacks (Right-Hand, 2025).

Companies must manage hundreds or thousands of agents in rapidly changing environments (Darktrace, 2025). The growing adoption of multi-agent systems will create new attack vectors and vulnerabilities if not secured from the outset (Weigand, 2025). Multi-agent AI systems bring security vulnerabilities, including data breaches, prompt injections, and data privacy risks (Weigand, 2025). The interconnectedness of multi-agent AI systems, though powerful, opens up new avenues for attacks. A breach in one AI agent could spread across the system, impacting various parts of the security infrastructure (Right-Hand, 2025).

Threats like data poisoning, prompt injection, and social engineering—already affecting single-agent systems—could pose even greater risks in multi-agent environments due to their expanded network of connections and interfaces, amplifying potential impacts (Weigand, 2025).

Perpetrators can exploit the novelty of agentic AI and victims' limited understanding of these systems. Insights into manipulative techniques used by various creatures to deceive their enemies are especially relevant in such situations. The "rare enemy syndrome," proposed by Dawkins (1982), offers a useful theoretical lens for understanding how victims can be easily deceived by new and unfamiliar cyber-threats or lures. The basic concept of rare enemy syndrome is that the enemy's manipulation is so uncommon that evolutionary development has not yet equipped the victim with an effective defense (De Jong, 2001). For example, perpetrators can exploit security weaknesses in Retrieval-Augmented Generation (RAG) and emerging techniques like prompt probing.

Retrieval augmented generation (RAG) AI, designed to enhance traditional LLMs by querying external data sources for real-time,

**Table 2**
Risk management framework in agentic AI.

| | Explanation | Context of agentic AI |
|---|---|---|
| Threat | Any potential risk that could compromise the security of an organization's systems, data, or network when agentic AI is involved. | *increased attack surface*<br>Rare enemy syndrome: Perpetrators can exploit security weaknesses unique to agentic AI such as those associated with RAG and prompt probing. |
| Vulnerability | Cyber-attack susceptibility | AI agents linked to other systems can increase vulnerabilities.<br>Subjective vulnerability: careless AI use leading to misperceived risk. |
| Potential impact | The possible impact or destruction that may occur when a vulnerability is exploited by a threat. | AI agents amplify risk impact by processing large volumes of personal data.<br>Autonomous vehicles using agentic AI could pose a risk of physical harm if hackers gain control. |

accurate responses, is gaining popularity in areas like financial analysis, healthcare, and e-commerce (Mistry, 2025). RAG enhances AI by combining information retrieval with LLMs. It queries external data sources, such as web pages and databases, and integrates relevant information into the LLM, enabling more context-aware and precise responses (TechMonitor., 2025). However, a typical RAG setup involves multiple components, including LLM-hosting platforms, open-source libraries, and vector databases that store externally sourced data. This reliance introduces significant security vulnerabilities. Research has found numerous unpatched CVEs in platforms like Ollama and misconfigured servers running llama.cpp and ChromaDB, exposing thousands of LLMs to potential exploitation. These security gaps provide opportunities for data theft, service manipulation, and sabotage. Since agentic AI shares many components with RAG, it faces similar risks (Mistry, 2025). Sandi Besen, an AI expert at IBM, stressed in her remarks that even the most advanced implementations of agentic RAG (Retrieval-Augmented Generation) cannot fully eliminate AI hallucinations, despite claims by some vendors (Woods, 2023).

Agentic RAG, already implemented in sectors like healthcare (VentureBeat, 2025), presents its own set of security challenges. Unlike conventional RAG, which simply retrieves external data, agentic RAG empowers AI agents to continuously refine queries, evaluate source reliability, and tailor responses to specific contexts (IBM, 2025). Agentic RAG utilizes intelligent agents to tackle complex questions, combining sequential reasoning and non-linguistic tools. These agents search, compare, and synthesize data from multiple documents, providing accurate, conclusive answers. The framework is scalable, with each set of new documents managed by a separate sub-agent (Akira AI, 2025). Agentic RAG enhances explainability by allowing traceability of AI actions, showing whether data came from document searches, web queries, or databases—crucial for compliance and accountability (IBM, 2025).

In traditional RAG workflows, operations occur within a closed system. However, with agents, AI autonomously interacts with external tools and data sources, raising concerns about data security and access control (IBM, 2025).

Prompt probing is a technique used to assess an agentic AI system's behavior, limitations, and vulnerabilities by testing it with crafted inputs. While researchers use it to understand AI responses, threat actors exploit it to uncover weaknesses for malicious activities like jailbreaking or prompt injection attacks. For instance, in an e-commerce setting, attackers might refine prompts to manipulate AI into approving fraudulent transactions, highlighting the risks of unauthorized system exploitation (SecurityWeek, 2025).

### 4.2. Vulnerability

Vulnerability is the extent to which an organization is susceptible to harm from cyber-attacks, with firms having weak cybersecurity practices being more likely to be affected than those with stronger practices (Kshetri, 2021). A cyberthreat could be exacerbated by vulnerabilities.

Vulnerability has two dimensions: objective and subjective (Busetta & Milito, 2009). Objective vulnerability relates to political, social, economic, and demographic factors that determine susceptibility to cyber-attacks, while subjective vulnerability involves an entity's self-perception of risk. Improving the accuracy of objective vulnerability assessments is crucial. Vulnerability analysis involves examining factors like third-party security mechanisms and the company's software and hardware. For example, a third-party service provider with weak cybersecurity can increase vulnerability by exposing the company's personally identifiable information (PII) (Kshetri, 2021).

Subjective vulnerability refers to an entity's self-perception of risk (Busetta & Milito, 2009), and careless use of AI systems, as reported by cybersecurity companies, can lead to a wrong perception of that risk (Trend, 2024). This underestimation of potential threats can leave organizations ill-prepared, increasing the likelihood that AI agents will act against the interests of their creators, users, or society, especially as these systems make decisions without full traceability.

The dynamic nature of AI agents makes effective threat detection challenging. These risks are further amplified when AI agents are connected to other systems, potentially increasing vulnerabilities (Ramesh, 2025). AI agents, as gateways to sensitive data and critical systems, create unique vulnerabilities that have higher impact in the event of an attack. For instance, if a logistics agent is compromised, it could spread malicious commands throughout the supply chain, leaving all supply chain partners vulnerable and causing significant and widespread disruption (Ramesh, 2025).

### 4.3. Potential impact

The impact of cyber-attacks includes identity theft, reputational damage, financial loss, and potential physical harm. For instance, compromised AI assistants could expose private organizational data, which could then be exploited for sophisticated social engineering and phishing attacks (SC World, 2025). Consequences are influenced by internal and external factors, such as the lack of cyber-liability insurance or strict laws on protecting PII (Kshetri, 2021).

The risks, including data exposure, unauthorized coding logic errors leading to breaches, and supply chain vulnerabilities from third-party libraries, can result in significant negative impact (Vella, 2024). Manipulation by external threats, including competitors, hackers, or cybercriminals, is a major risk to agentic AI systems. Such interference can reduce algorithm accuracy or cause it to go rogue, leading to financial losses and reputational damage. Even with a small role in a department's operations, a cyberattack could have severe consequences. A typical 2 % failure rate might be manageable, but if it spikes to 100 %, the resulting downtime could trigger expensive, large-scale failures (Security Boulevard, 2025).

The impact of these risks is amplified as AI agents process large volumes of personal data. For example, AI agents managing emails or investment decisions handle sensitive personal or financial information, increasing a company's vulnerability to cyberattacks (Dorsey & Whitney, 2025). Similarly, agents handling tasks like trip planning or email responses require access to personal data, such

as schedules, preferences, financial details, and communications (SC World, 2025). This extensive data collection increases the risk of overreach, potentially transforming these systems from assistants into surveillance tools that monitor and profile users, raising concerns about privacy and misuse of sensitive information (SC World, 2025).

Cyber-attacks can also result in potential physical harm, especially if hackers gain control of autonomous vehicles using agentic AI (SC World, 2025). Such breaches can have serious consequences, including jeopardizing public safety.

The impact can be more significant if the firm's jurisdiction enforces strict laws on failing to protect PII. For instance, agentic AI relies heavily on data, increasing risks under privacy frameworks like GDPR or CCPA (Above the Law, 2025).

## 5. The malicious potential of agentic AI in cybercrime

While threat actors are experimenting with autonomous attack agents, their practical use in large-scale attacks remains distant (Ranger, 2025). In contrast, agentic AI is already empowering defenders with predictive threat detection and automated response systems, creating a structural advantage. Cybercriminals, hindered by underdeveloped AI capabilities, remain reliant on costly, reactive strategies that struggle against AI-enhanced defenses. Traditional attack methods are increasingly ineffective, and newer tactics require advanced tools or risky techniques with high failure rates—a barrier for less-resourced groups. These challenges drive up operational costs, rendering conventional cybercrime models unsustainable (KnowBe4, 2025).

Cybercriminals are nonetheless leveraging agentic AI for malicious purposes, such as self-improving phishing campaigns. The asymmetric nature of cyberattacks enables low-resource actors to target high-value assets (Masters, 2011; Lavorgna, 2015). For example, poisoning just 0.01 % of massive datasets like LAION-400M or COYO-700M in 2023 cost attackers as little as $60—a minimal investment with disproportionate impact (Carlini et al., 2024, May 6; Flynn, 2024). Malwarebytes predicts agentic AI could mark a paradigm shift, enabling attackers to autonomously strategize, reason, and execute multi-stage operations (Klappholz, 2025). These systems can identify vulnerabilities, adapt tactics in real-time, and combine communication channels (e.g., deepfake calls with phishing emails) to bypass defenses (Sjouwerman, 2025).

This dynamic may shift as AI agents mature, raising cybersecurity concerns. These tools could soon enable cybercriminals to automate labor-intensive tasks like vulnerability scanning, target selection, and attack execution, bypassing the need for specialized human expertise. Stockley emphasizes that AI's scalability and cost-efficiency will likely make it the primary driver of future attacks, particularly in ransomware (Heaven, 2025). By delegating critical functions to AI, attackers could exponentially expand operations, overcoming current limitations tied to human labor (Klappholz, 2025).

Agentic AI also amplifies systemic risks. Compromising a single component—such as poisoned training data or orchestration layer flaws—can corrupt algorithms, trigger denial-of-service attacks, or enable lateral movement for data exfiltration (Baker, 2025; Security Boulevard, 2025). Biased or inaccurate data can amplify hallucinations or impair decision-making, leading to unintended behaviors that go undetected without human oversight (SACA, 2014).

Synthetic identity fraud exemplifies AI's evolving threat landscape. While this tactic predates generative AI—as seen in the 2017 case where fraudsters blended fake and real data to open fraudulent accounts (U.S. Immigration and Customs Enforcement, 2022)—agentic AI could automate and scale these schemes. In 2024, IDology reported a 45 % surge in synthetic fraud among fintechs, driven by AI's ability to generate convincing identities and evade detection (PRNewswire, 2024). By automating account creation and adaptive evasion tactics, AI eliminates reliance on human intermediaries, making fraud faster and harder to trace.

Despite these advancements, cybercriminals face hurdles. Current autonomous attack agents remain unreliable, requiring significant refinement for large-scale deployment (Klappholz, 2025). As ransomware gangs and malvertising campaigns adopt AI-driven automation, defenders must prepare for a future where attacks are not only faster and cheaper but also *self-optimizing*.

## 6. Discussion and implications

Let us start this section with the research questions that we wanted to address. Regarding RQ1, agentic AI can transform cybersecurity practices by automating and enhancing threat detection, response, and mitigation. By leveraging machine learning and adaptive algorithms, agentic AI systems can continuously monitor network traffic, identify suspicious patterns, and predict potential vulnerabilities before they are exploited. These systems can autonomously adapt to new threats in real-time, providing a more dynamic defense mechanism that outpaces traditional, human-driven approaches. Additionally, agentic AI can be deployed in incident response, making immediate decisions and executing predefined actions to neutralize threats, thereby reducing the impact of attacks and improving the overall efficiency of cybersecurity operations.

As to RQ2, the integration of agentic AI also introduces significant cybersecurity risks. One major concern is the potential for adversarial manipulation, where malicious actors might exploit weaknesses in AI models to bypass security protocols or manipulate decision-making processes. Furthermore, as AI systems become more autonomous, there is a risk of loss of oversight, where the AI might make decisions that are not aligned with organizational goals or ethical considerations. Additionally, vulnerabilities in AI models themselves could be targeted, leading to data breaches or the exploitation of AI systems for nefarious purposes. Proper safeguards, continuous monitoring, and transparent AI design are essential to mitigate these risks.

### *6.1. The agentic AI equation: costs, benefits, and governance*

In 2023, the median annual wage for information security analysts was $120,360 (U.S. Bureau of Labor Statistics, 2024). With such high labor costs, AI agents offer a compelling advantage by performing security tasks with greater speed and accuracy. However,

developing these agents remains a highly complex undertaking. Most organizations lack the in-house expertise needed to build and deploy AI agents effectively. As a result, many will turn to external consultants or vendor-provided solutions, while more strategic firms will form partnerships to integrate cutting-edge agent technologies. Even when companies succeed in developing in-house AI agents, they often underestimate the long-term costs—ongoing maintenance, monitoring, and optimization frequently exceed initial development expenses. According to Forrester's 2025 AI predictions, 75 % of in-house AI agent projects are expected to fail (Chaurasia & Maheshwari, 2024).

Enterprises are already integrating AI agent capabilities into platforms like Microsoft Copilot Studio and AWS Bedrock (Vella, 2024). The above discussion also makes it clear that while AI agents offer new capabilities, they also introduce additional risks. A key advantage of AI agents is their ability to discover, communicate, and collaborate with other agents. However, without well-defined communication boundaries and explicit permissions, this poses significant data privacy risks—challenges that traditional application testing alone cannot resolve (Weigand, 2025).

Proactive business and IT leaders will embrace agentic AI while mitigating risks by setting clear guardrails, enforcing strict data access policies, and promoting organizational best practices (Weigand, 2025). It is essential to understand that AI agents will operate in organizations like human employees, interacting with other agents and integrating into HR systems with distinct permissions and access (Weigand, 2025). Security in agentic AI must prioritize mitigating risks inherent in agent onboarding/offboarding while ensuring data quality, governance, and resilience against autonomous threats (Gomes, 2024). Having noted the potential for criminal use, cybercriminals are experimenting with AI-driven attacks, though experts suggest refinement is needed before they can be fully effective (Weigand, 2025).

A gap continues to exist between LLM assistants and fully autonomous agents (Vella, 2024). Closing this gap requires strong governance to ensure security and trust, starting with narrow use cases before broader adoption (Vella, 2024). As AI agents grow more advanced and interconnected, they may introduce additional security vulnerabilities and potential data leaks. If not carefully implemented, agentic AI could pose privacy and cybersecurity risks to sensitive data (Weigand, 2025). Thus, a governance framework that aligns with national and global standards is critical for mitigating cybersecurity risks tied to agentic AI. These policies ensure that implementation and use remain ethical, transparent, and secure, addressing autonomy-related threats. Security professionals must also consider the level of agency granted to the system, its complexity, and the environment in which it operates, as these factors significantly impact its potential security risks (Security Boulevard., 2025). Implementing threat modeling, least-privilege access, and separating trusted from untrusted zones can reduce vulnerabilities. Additionally, monitoring agent activity and maintaining audit trails help detect anomalies effectively (Ramesh, 2025).

Firms facing higher risks should implement information sharing, develop threat-specific policies, and enhance cybersecurity awareness through training to minimize impacts. They must also obtain consent for data collection, ensure compliance with sector-specific privacy laws (e.g., HIPAA), and provide oversight to prevent legal and security risks (Brown, 2025). Compliance with privacy regulations, such as the California Consumer Privacy Act, is essential, including offering opt-out options for automated decision-making and disclosing data practices. Without proper oversight, AI agents may engage in scams, create vulnerable software, or cause security breaches, making human supervision and real-time monitoring crucial (Dorsey & Whitney, 2025). To mitigate risks in multi-agent AI systems, it is essential to implement strategies that isolate and contain breaches within these complex networks, preventing them from cascading throughout the system (Right-Hand AI., 2025a,b).

A key approach to securing automation is restricting an agent's level of autonomy based on risk. If an agent cannot be fully assessed for vulnerabilities or trusted, its capabilities should be limited until stronger security measures are in place (Baker, 2025). The rise of agentic AI may also necessitate modifications in existing policies, such as Bring Your Own Device (BYOD) protocols. As AI systems become more integrated into corporate networks and data flows, the risks of personal devices accessing these systems may outweigh the benefits of BYOD. CISOs may need to enforce stricter controls or switch to corporate-managed devices to reduce the risk of data leaks and unauthorized AI tool usage (Right-Hand AI., 2025a,b).

Beyond technical safeguards, agentic AI also raises complex policy and regulatory challenges that must be addressed in parallel. Agentic AI depends on access to extensive, high-quality, and diverse datasets to achieve its full capabilities. However, sectors such as healthcare, finance, and national security face strict privacy laws and compliance requirements that restrict how data is used or shared. Privacy-Enhancing Technologies (PETs) mitigate this conflict by allowing AI systems to analyze sensitive data without exposing it directly. Cross-border data governance introduces additional complexities, particularly when agentic AI must operate across jurisdictions with differing regulations. Resolving policy challenges—including mandates, liability frameworks, and international data-sharing restrictions—is essential to ensure agentic AI remains effective, trustworthy, and compliant with legal standards (Moran, 2025).

There is also the need for employee training, and automated detection systems to mitigate these risks (Weigand, 2025). Enhancing security awareness through dynamic training tools tailored to risk levels is essential. Organizations should also educate employees about social engineering risks and foster a culture of cybersecurity (Sjouwerman, 2025). To mitigate AI agents' potential threats, IT leaders should educate their organizations on AI agent risks, detect and flag anomalous activities, and map all AI agent interactions to ensure compliance with enterprise policies (Vella, 2024). To combat advanced social engineering, organizations can also use agentic AI to monitor and detect threats, analyze behaviors, and prioritize vulnerabilities. As AI evolves, employees must be prepared for autonomous agents in the workplace, and businesses should deploy their own AI-based security agents to stay ahead of cybercriminals (Sjouwerman, 2025).

A multi-layered security approach, consisting of predictive generative agentic AI systems, is essential for safeguarding against the potential cyber-threats. Generative AI providers like OpenAI, Anthropic, and Google implement content safeguards, while enterprises add their own AI gateways and protections. Agentic software requires internal guardrails, filtering prompts, and monitoring responses.

As multi-agent systems emerge, additional safeguards will be needed to manage inter-agent interactions (Baker, 2025). For security, businesses using agent or multi-agent systems have options. Guardrails can be applied to inputs and outputs, and limiting an agent's actions makes it easier to secure than a complex LLM. Multi-agent systems with well-defined roles are easier to secure than single agents. Since prompt injection techniques evolve rapidly, relying solely on LLM guardrails is insufficient. Agent security will likely require a mix of rules and monitoring for the LLM and policy enforcement for the actions (Baker, 2025).

Deploying agentic AI and AI agents for cybersecurity or any other purposes is also a resource-intensive, time-consuming, and complex process. Agents can consume CPU, memory, and network resources, potentially impacting performance, particularly in environments with high workloads or limited resources. Managing hundreds or thousands of agents in highly dynamic environments, such as containers or serverless functions, can be complex and require significant effort. Installing agents on each workload or instance can be time-consuming, especially in large or complex environments (Darktrace., 2025).

The rise of AI agents does not diminish the usefulness of agentless systems. A combined approach utilizes agent-based solutions for deep protection and monitoring of critical systems, and agentless solutions for fast deployment, scalability, and visibility in dynamic cloud environments. For example, a financial services company may use agent-based security for sensitive databases and agentless solutions for transient cloud workloads, balancing protection with performance and scalability (Darktrace, 2025).

To fully realize the benefits of AI agents while mitigating the high costs and complexity, organizations must embed these strategic considerations into formal policy frameworks. Cyber AI strategies should include clear governance structures that define roles, responsibilities, and escalation paths when AI systems make autonomous decisions. Procurement policies should evaluate AI vendors not only on technical capabilities but also on commitments to ethical use, transparency, and data protection. Additionally, budgeting and planning policies must account for long-term AI lifecycle costs—including training, updates, performance monitoring, and auditing—to avoid underinvestment in critical phases post-deployment. By aligning these operational, ethical, and financial aspects within cohesive policies, organizations can ensure that the adoption of AI agents strengthens—not complicates—their cybersecurity posture (Tortora Brayda, 2025).

### 6.2. *Future research*

Before concluding, we propose several promising directions for future research. First, one issue raised above is that the use of autonomous AI agents by cybercriminals for attacks is still hindered by their unreliability. Experts expect these systems to undergo significant improvements before they become viable tools (Klappholz, 2025), and this situation is expected to change. Prior research has shown that the world's major cybercrime firms have rapidly adopted new technologies and business models (Goodman, 2011; Kshetri, 2010, 2013b). IT security firms develop new defenses against emerging threats, while cybercriminals continually enhance tactics to evade security measures (Kraemer-Mbula et al., 2013). Prior research has highlighted that cybercrime organizations closely mirror the business models of legitimate corporations (Goodman, 2011; Kshetri, 2010, 2013a, 2013b). Future research could explore how the rapid evolution of autonomous AI agents in cybercrime may change the dynamics of cybercriminal organizations. Given that these organizations often mirror the business models of legitimate corporations, understanding how they adopt and adapt AI technologies will be crucial in anticipating new threats. Additionally, research could focus on how IT security firms and cybercriminals engage in a constant arms race, with security measures evolving to counteract increasingly sophisticated tactics.

Cybercriminals have adopted increasingly sophisticated technologies and business strategies in their operations, including Ransomware-as-a-Business (RaaB) models that mirror corporate structures and leverage market dynamics to optimize attacks (Kshetri & Voas, 2022). Yet despite these advances, reliable real-world deployment of autonomous attack agents remains years away. In the meantime, agentic AI gives defenders a critical structural advantage through early, sustained innovation in predictive analytics and autonomous response. Adversaries now face an asymmetric landscape: lagging development of agentic tools forces reliance on reactive and costly tactics to evade AI-powered defenses. Legacy attack vectors are collapsing under the weight of AI's predictive precision, while untested, adaptive approaches require high-risk investments with uncertain returns. As a result, less-resourced groups are falling further behind, and growing operational costs are eroding the profitability of conventional cybercrime—accelerating the obsolescence of outdated playbooks. To stay ahead, researchers must now focus on the technical and strategic thresholds at which agentic AI will enable cyberattacks with true autonomy, from network infiltration to exfiltration, without human input. Understanding this transition is key to proactively securing systems against the next era of AI-driven threats.

The asymmetric nature of cyberattacks allows actors with limited financial and technical resources to compromise high-value targets (Masters, 2011; Lavorgna, 2015). Future research could examine whether this asymmetry has shifted in the context of agentic AI, particularly in terms of the advantages cybercriminals and other offenders hold over defenders. Understanding how agentic AI alters the balance between attackers and defenders could offer valuable insights into emerging cybersecurity challenges.

Another intriguing avenue for future research is to apply principal-agency theory in the context of cybercrime networks. This theory explores the relationship in which a principal delegates authority to an agent to perform tasks or make decisions on their behalf (Eisenhardt, 1989; Kiser, 1999). The agent's actions are intended to serve the principal's interests, often in the role of a representative or employee (Mitnick, 1973; Soudijn & Zhang, 2016). Prior research in the context of conventional crimes has suggested that criminal enterprises face unique challenges, including lack of legal contract enforcement, law enforcement interference, and potential violence from rivals. These uncertainties increase disputes and raise questions about how criminal leaders manage agents and resolve conflicts (Soudijn & Zhang, 2016). When tasks are outsourced or assigned, as seen in organized criminal activities, agency problems and associated costs inevitably arise. Applying principal-agency theory provides insights into the daily operations of illicit enterprises and the interactions among their members. (Soudijn & Zhang, 2016). Future research could examine how cybercriminals leverage AI agents to automate attacks, evade detection, and optimize illicit operations. Unlike human operatives, AI-driven agents lack personal

motives but require precise programming to align with criminal objectives while avoiding detection. This shift presents new challenges in maintaining control, ensuring task completion, and mitigating risks. Applying agency theory can offer insights into how criminals manage AI agents, adapt to evolving security measures, and address issues arising from AI errors or unintended behaviors.

## 7. Conclusion

Agentic AI represents a transformative advancement in cybersecurity by enhancing threat detection, autonomous response, and operational efficiency. Companies like ReliaQuest, CrowdStrike, and Twine are already leveraging this technology to streamline SOC tasks, improve accuracy, and reduce response times. Moreover, agentic AI's ability to continuously analyze and adapt throughout the SDLC offers proactive protection against vulnerabilities. However, these benefits are accompanied by significant risks, such as expanded attack surfaces, data leaks, and potential manipulation by cybercriminals. The dynamic nature of multi-agent systems and their interactions with sensitive data further amplifies security challenges. As adoption grows, robust monitoring, governance, and risk mitigation strategies are crucial to maximize agentic AI's potential while minimizing vulnerabilities. Balancing innovation with cybersecurity resilience will be essential for navigating the future of agent-driven solutions.

## References


Above the Law. (2025). Agents vs. agentic AI: What in-house counsel need to know about these 2 AI frontiers. Retrieved from https://abovethelaw.com/2025/02/agents-vs-agentic-ai-what-in-house-counsel-need-to-know-about-these-2-ai-frontiers.

Akira, A. I. (2025). RAG vs. agentic RAG. Retrieved from https://www.akira.ai/blog/rag-vs.-agentic-rag.

Baker, B. (2025). Agentic AI paves the way for sophisticated cyberattacks. *AI Business*. from https://aibusiness.com/generative-ai/agentic-ai-paves-the-way-for-sophisticated-cyberattacks. (Accessed 24 February 2025).

Barker, S. (2025). *Microsoft security Copilot Agents unveiled.* the register. https://www.theregister.com/2025/03/24/microsoft_security_copilot_agents/.

Boston Consulting Group. (2025). AI agents. Retrieved January 18, 2025, from https://www.bcg.com/capabilities/artificial-intelligence/ai-agents.

Brown, A. (2025). Agents vs agentic AI: What in-house counsel need to know about these 2 AI frontiers. *Above the Law*. Retrieved from https://abovethelaw.com/2025/02/agents-vs-agentic-ai-what-in-house-counsel-need-to-know-about-these-2-ai-frontiers.

Busetta, A., & Milito, A. M. (2009). Socio-demographic vulnerability: The condition of Italian young people. *Social Indicators Research, 97*(3), 375–396.

Carlini, N., Jagielski, M., Choquette-Choo, C. A., Paleka, D., Pearce, W., Anderson, H., Terzis, A., Thomas, K., & Tramèr, F. (2024). Poisoning web-scale training datasets is practical (arXiv:2302.10149v2). *arXiv [Preprint]* https://arxiv.org/html/2302.10149v2.

Carter, S. (2025). Three AI agents built on blockchain to transform crypto, DeFi, gaming. *Forbes*. https://www.forbes.com/sites/digital-assets/2025/02/06/three-ai-agents-built-on-blockchain-to-transform-crypto-defi-gaming/.

Castellanos, S. (2021). *Agentic AI: Promising use cases for business*. CIO. Retrieved February 24, 2025, from https://www.cio.com/article/3603856/agentic-ai-promising-use-cases-for-business.html.

Chaurasia, J., & Maheshwari, S. (2024). Predictions 2025: Artificial intelligence. *Forrester*. https://www.forrester.com/report/predictions-2025-artificial-intelligence/RES181360.

Chiodi, M. (2025). Agentic AI in cybersecurity: Closing the last mile in identity security. *Cerby*. Retrieved February 24, 2025, from https://www.cerby.com/resources/blog/agentic-ai-in-cybersecurity.

CIO. (2024). *Agentic AI: 6 promising use cases for business*. CIO. Retrieved February 24, 2025, from https://www.cio.com/article/3603856/agentic-ai-promising-use-cases-for-business.html.

Columbus, L. (2025). Cybersecurity at AI speed: How agentic AI is supercharging SOC teams in 2025. *VentureBeat*. Retrieved February 24, 2025, from https://venturebeat.com/ai/cybersecurity-at-ai-speed-agentic-ai-supercharging-soc/.

Cosive. CrowdStrike Falcon agent: A radical new approach proven to stop breaches. Cosive, [Online]. Available: https://www.cosive.com/capabilities/crowdstrike-falcon-agent#:~:text=The%20CrowdStrike%20Falcon%C2%AE%20platform%20is%20built%20on,efficacy%20against%20a%20wide%20variety%20of%20threats.&text=A%20single%20lightweight%20agent%20works%20everywhere%2C%20including,providing%20protection%20even%20when%20endpoints%20are%20offline. (Accessed 13 February 2025).

CrowdStrike. (2025a). CrowdStrike delivers the next breakthrough in AI-powered agentic cybersecurity with Charlotte AI detection triage. *Business Wire*. Retrieved February 24, 2025, from https://www.businesswire.com/news/home/20250212935384/en/CrowdStrike-Delivers-the-Next-Breakthrough-in-AI-Powered-Agentic-Cybersecurity-with-Charlotte-AI-Detection-Triage.

CrowdStrike. (2025b). CrowdStrike announces availability of agentic AI Charlotte AI detection triage. *Intelligent CIO Africa*. https://www.intelligentcio.com/africa/2025/03/21/crowdstrike-announces-availability-of-agentic-ai-charlotte-ai-detection-triage/.

CyberArk. (n.d.). What is agentic AI and AI agents? https://www.cyberark.com/what-is/agentic-ai-and-ai-agents/.

Darktrace. (2025). Agent vs agentless cloud security: Why deployment methods matter. https://darktrace.com/blog/agent-vs-agentless-cloud-security-why-deployment-methods-matter. Retrieved from.

Darktrace. (2024a). Darktrace endpoint solution brief. Retrieved from https://cdn.prod.website-files.com/626ff4d25aca2edf4325ff97/66cdb28d6944269cc68d70c5_Darktrace%20Endpoint%20Solution%20Brief.pdf.

Darktrace. (2024b). Darktrace endpoint. Retrieved from https://darktrace.com/products/endpoint.

Darktrace. (2025). Agent vs. agentless cloud security: *Why deployment methods matter*. Retrieved from https://darktrace.com/blog/agent-vs-agentless-cloud-security-why-deployment-methods-matter#:~:text=This%20integrated%20strategy%20ensures%20that,new%20threats%20and%20regulatory%20requirements.&text=Powered%20by%20multilayered%20AI%2C%20Darktrace,environments%20in%20the%20Solution%20Brief.

Dawkins, R. (1982). *The extended phenotype*. New York: Oxford University Press.

De Jong, W. M. (2001). Manipulative tactics in budgetary games: The art and craft of getting the money you don't deserve. *Knowledge, Technology & Policy, 14*(1), 50–66.

Dorsey & Whitney, L. L. P. (2025). Launching agentic AI in an uncertain U.S. regulatory landscape. Retrieved February 24, 2025, from *JD Supra* https://www.jdsupra.com/legalnews/launching-agentic-ai-in-an-uncertain-u-2524341.

Eisenhardt, K. M. (1989). Agency theory: An assessment and review. *Academy of Management Review, 14*(1), 57–74. https://doi.org/10.2307/258191

Finch, B. (2014). Why cybersecurity must be defined by process, not tech. *The Wall Street Journal*. Retrieved from http://blogs.wsj.com/cio/2014/12/11/why-cybersecurity-must-be-defined-by-process-not-tech/.

Flynn, S. (2024). *Data poisoning: How AI models are corrupted*. ReHack. https://rehack.com/ai/machine-learning/data-poisoning/.

Gomes, G. (2024). Agentic AI breakthrough in tech. *CTO Magazine*. https://ctomagazine.com/agentic-ai-breakthrough-in-tech/.

Goodman, M. (2011). What business can learn from organized crime. *Harvard Business Review, 89*(11), 27–30.

Guerrero, C. (2025). The power of agentic AI: How autonomous agents are revolutionizing cybersecurity and application security. *Dev.to*. Retrieved from https://dev.to/friendgrass7/the-power-of-agentic-ai-how-autonomous-agents-are-revolutionizing-cybersecurity-and-application-h31.

Heaven, W. D. (2025). Cyberattacks by AI agents are coming. *MIT Technology Review*. https://www.technologyreview.com/2025/04/04/1114228/cyberattacks-by-ai-agents-are-coming/.

High, M. (2025). Nvidia and CrowdStrike push agentic AI in cybersecurity. *Cyber Magazine*. https://cybermagazine.com/articles/nvidia-and-crowdstrike-push-ai-driven-cybersecurity.
Hodjat, B. (2025). *How to ensure the safety of modern AI agents and multi-agent systems*. World Economic Forum. Retrieved https://www.weforum.org/stories/2025/01/ai-agents-multi-agent-systems-safety/. (Accessed 24 February 2025).
Hoque, F. (2025). How agentic AI will shape the future of business. *Fast ompany*. https://www.fastcompany.com/91282782/how-agentic-ai-will-shape-the-future-of-business.
IBM. (2025). AI detectives and agentic RAG. Retrieved from https://www.ibm.com/think/news/ai-detectives-agentic-rag.
Institute for Experiential AI. (2025). *Agentic AI: Institute for experiential AI position*. Northeastern University. Retrieved February 24, 2025, from https://ai.northeastern.edu/news/agentic-ai-institute-for-experiential-ai-position.
Kaufman, S. (2024). *Beyond ChatGPT: The rise of agentic AI and its implications for security*. CSO Online. Retrieved February 24, 2025, from https://www.csoonline.com/article/3574697/beyond-chatgpt-the-rise-of-agentic-ai-and-its-implications-for-security.html.
Kiser, E. (1999). Comparing varieties of agency theory in economics, political science, and sociology: An illustration from state policy implementation. *Sociological Theory, 17*(2), 146–170. https://doi.org/10.1111/0735-2751.00073
Klappholz, S. (2025). *Agentic AI could be a blessing and a curse for cybersecurity*. ITPro. Retrieved February 24, 2025, from https://www.itpro.com/security/cyber-crime/agentic-ai-cybersecurity-risks.
KnowBe4. (2025). *Emergent agentic AI defense*. KnowBe4 Blog. Retrieved April 18, 2025, from https://blog.knowbe4.com/emergent-agentic-ai-defense.
Kraemer-Mbula, E., Tang, P., & Rush, H. (2013). The cybercrime ecosystem: Online innovation in the shadows? *Technological Forecasting and Social Change, 80*(3), 541–555. https://doi.org/10.1016/j.techfore.2012.07.017
Kshetri, N. (2010). *The global cyber-crime industry: Economic, institutional and strategic perspectives*. Springer-Verlag.
Kshetri, N. (2013a). Cybercrimes in the former soviet union and central and eastern Europe: Current status and key drivers. *Crime, Law and Social Change, 60*(1), 39–65. https://doi.org/10.1007/s10611-013-9446-7
Kshetri, N. (2013b). *Cybercrime and cybersecurity in the global South*. Palgrave Macmillan.
Kshetri, N. (2021). *Cybersecurity management: An organizational and strategic approach*. University of Toronto Press.
Kshetri, N., & Voas, J. (2022). Ransomware as a business (RaaB). *IT Professional, 24*(2), 83–87. https://doi.org/10.1109/MITP.2022.3157208
Lavorgna, A. (2015). Organised crime goes online: Realities and challenges. *Journal of Money Laundering Control, 18*(2), 153–168. https://doi.org/10.1108/JMLC-10-2014-0035
Lisowski, E. (2024). AI agents vs agentic AI: What's the difference and why does it matter? *The Medium*. Retrieved December 29, 2024, from https://medium.com/@elisowski/ai-agents-vs-agentic-ai-whats-the-difference-and-why-does-it-matter-03159ee8c2b4.
Mistry, B. (2025). *Looking before we leap: Why security is essential to agentic AI success*. TechRadar. https://www.techradar.com/pro/looking-before-we-leap-why-security-is-essential-to-agentic-ai-success.
Mitnick, B. M. (1973). Fiduciary rationality and public policy: The theory of agency and some consequences. In *Paper presented at the 1973 annual meeting of the American political science association*. New Orleans: LA. In Proceedings of the APSA, 1973.
Moran, O. (2025). *Unlocking the potential of agentic AI with privacy-enhancing technologies*. DualityTech. https://dualitytech.com/blog/unlocking-the-potential-of-agentic-ai-with-privacy-enhancing-technologies/.
National Institute of Standards and Technology (NIST). (2024). Risk management framework guidance (NIST SP 800-37r2). Retrieved from https://www.electri.org/wp-content/uploads/2024/03/NIST.SP_.800-37r2-Risk-Management-Framework-Guidance.pdf.
Nawaz, S. (2024). Agentic AI vs AI agents: A detailed comparison. *Ampcome*. Retrieved February 24, 2025, from https://www.ampcome.com/post/agentic-ai-vs-ai-agents-a-detailed-comparison.
Precedence Research. (2024). Artificial intelligence (AI) in cybersecurity market size, share, and trends 2024 to 2034. *Precedence Research*. Retrieved February 24, 2025, from https://www.precedenceresearch.com/artificial-intelligence-in-cybersecurity-market.
Press, G. (2024). This AI agent will defend you from cyber attacks. *Forbes*. Retrieved from https://www.forbes.com/sites/gilpress/2024/11/20/this-ai-agent-will-defend-you-from-cyber-attacks/.
PRNewswire. (2024). Synthetic identities and deep fakes: The new face of fraud with Gen AI. *Machine Learning Analytics News*. https://aithority.com/machine-learning/synthetic-identities-and-deep-fakes-the-new-face-of-fraud-with-gen-ai/.
Ramesh, R. (2025). What enterprises need to know about agentic AI risks: Mitigating cybersecurity, privacy risks for new class of autonomous agents. *BankInfoSecurity*. Retrieved February 24, 2025, from https://www.bankinfosecurity.com/what-enterprises-need-to-know-about-agentic-ai-risks-a-27282.
Ranger, S. (2025). Agentic AI might take years to transform security, but cyber defenders must prepare now. *BetaNews*. https://betanews.com/2025/03/30/agentic-ai-might-take-years-to-transform-security-but-cyber-defenders-must-prepare-now/.
Red Hat. (2024). *What is agentic AI?* Red Hat. Retrieved January 15, 2025, from https://www.redhat.com/en/topics/ai/what-is-agentic-ai.
Right-Hand AI. (2025a). Agentic AI in cybersecurity. Retrieved from https://right-hand.ai/blog/agentic-ai-in-cybersecurity/.
Right-Hand AI. (2025b). Adopting agentic AI in cybersecurity: What CISOs expect in 2025. Retrieved February 24, 2025, from https://right-hand.ai/blog/agentic-ai-in-cybersecurity/.
Riley, D. (2024). AI meets cybersecurity: Twine launches with $12M funding for digital cyber employees. *SecurityWeek*. Retrieved from https://siliconangle.com/2024/11/20/ai-meets-cybersecurity-twine-launches-12m-funding-digital-cyber-employees//.
SACA. (2014). Generating value from big data analytics. *Information Systems Audit and Control Association* [White paper] http://www.isaca.org/Knowledge-Center/Research/ResearchDeliverables/Pages/Generating-Value-From-Big-Data-Analytics.aspx.
SC World. (2025). Five privacy concerns around agentic AI. Retrieved from https://www.scworld.com/perspective/five-privacy-concerns-around-agentic-ai.
Security Boulevard. (2025). Developing security protocols for agentic AI applications. *Security Boulevard*. https://securityboulevard.com/2025/01/developing-security-protocols-for-agentic-ai-applications/.
SecurityWeek. (2025). How hackers manipulate agentic AI with prompt engineering. Retrieved from https://www.securityweek.com/how-hackers-manipulate-agentic-ai-with-prompt-engineering/.
Sjouwerman, S. (2025). Code Property Graph - Specification and Tooling [GitHub repository]. GitHub. https://github.com/ShiftLeftSecurity/codepropertygraph. (Accessed 9 February 2025).
Sjouwerman, S. (2025). How agentic AI will be weaponized for social engineering attacks. *SecurityWeek*. https://www.securityweek.com/how-agentic-ai-will-be-weaponized-for-social-engineering-attacks.
Soudijn, M. R. J., & Zhang, S. X. (2016). Principal-agency theory in illegal markets: Cooperation and conflict among Chinese loansharks. *Asian Criminology, 11*(3), 231–247. https://doi.org/10.1007/s11417-015-9229-7
TechMonitor. (2025). Agentic AI cyber risks. Retrieved from https://www.techmonitor.ai/comment-2/agentic-ai-cyber-risks/?cf-view.
The Hacker News. (2025). *What is attack surface management?* The hacker news. https://thehackernews.com/2025/02/what-is-attack-surface-management.html.
Tortora Brayda, C. (2025). Agentic AI Vs. AI agents: Shaping the future of cybersecurity. *Forbes*. https://www.forbes.com/councils/forbestechcouncil/2025/04/14/agentic-ai-vs-ai-agents-shaping-the-future-of-cybersecurity/.
Trend, M. (2024). Agentic AI takeover. Retrieved from https://www.trendmicro.com/en_us/research/24/h/agentic-ai-takeover.html.
U.S. Bureau of Labor Statistics. (2024). *Information security analysts*. Occupational outlook handbook. https://www.bls.gov/ooh/computer-and-information-technology/information-security-analysts.htm.
U.S. Immigration and Customs Enforcement. (2022). HSI investigates synthetic identities scheme that defrauded banks nearly $2M. https://www.ice.gov/news/releases/hsi-investigates-synthetic-identities-scheme-defrauded-banks-nearly-2m#:~:text=Cato%20and%20his%20co%2Dconspirators,as%20their%20social%20security%20numbers.
Vella, H. (2024). *Agentic AI set to rise, with new cybersecurity risks: Gartner*. AI Business. https://aibusiness.com/automation/agentic-ai-set-to-rise-with-new-cybersecurity-risks-gartner.

VentureBeat. (2025). Medical training's AI leap: How agentic RAG, open-weight LLMs, and real-time case insights are shaping a new generation of doctors at NYU Langone. Retrieved from https://venturebeat.com/ai/medical-trainings-ai-leap-how-agentic-rag-open-weight-llms-and-real-time-case-insights-are-shaping-a-new-generation-of-doctors-at-nyu-langone/.

Weigand, S. (2025). *Cybersecurity in 2025: Agentic AI to change enterprise security and business operations in year ahead*. SC World. Retrieved February 24, 2025, from https://www.scworld.com/feature/ai-to-change-enterprise-security-and-business-operations-in-2025.

Williams, S. (2025). ReliaQuest's AI agent boosts cybersecurity precision. *SecurityBrief Australia*. Retrieved https://securitybrief.com.au/story/reliaquest-s-ai-agent-boosts-cybersecurity-precision. (Accessed 24 February 2025).

Woods, B. (2023). *Agentic RAG AI: More marketing hype than tech advance?* ComputerWorld. https://www.computerworld.com/article/3487242/agentic-rag-ai-more-marketing-hype-than-tech-advance.html.

World Economic Forum. (2024). *Strategic cybersecurity talent framework*. World Economic Forum. from https://www.weforum.org/publications/strategic-cybersecurity-talent-framework/. (Accessed 16 February 2025).